

CSE4261: Neural Network and Deep Learning

Assignment-9

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1. 2D Feature Representation Using Autoencoder

We trained a convolutional autoencoder on 5000 CIFAR-10 images. The encoder was modified to project the latent space into 2D. After 20 epochs, the training loss converged around 0.037, with validation loss around 0.0359, indicating stable learning.

Analysis: The learned 2D features allowed visual separation among some classes. However, due to low-dimensional bottleneck, class boundaries overlapped in some regions.

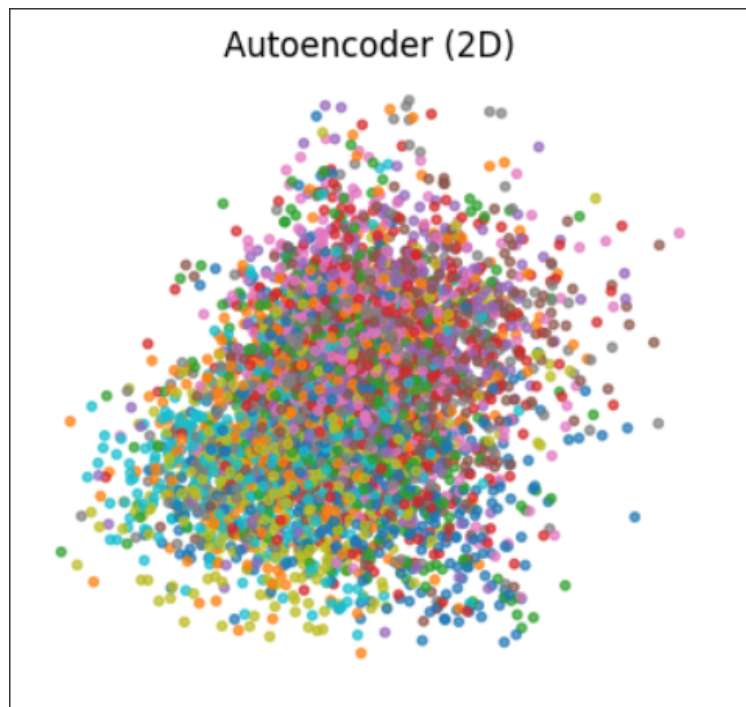


Figure 1: 2D visualization of encoded CIFAR-10 features

2. Comparison with Pre-trained CNN Features (PCA, t-SNE)

We extracted high-level features using MobileNetV2 (pre-trained on ImageNet), then applied PCA and t-SNE on MobileNet.

Findings:

- MobileNet features clustered more clearly in both PCA and t-SNE, showing better separability of classes.
- Autoencoder features were less discriminative.

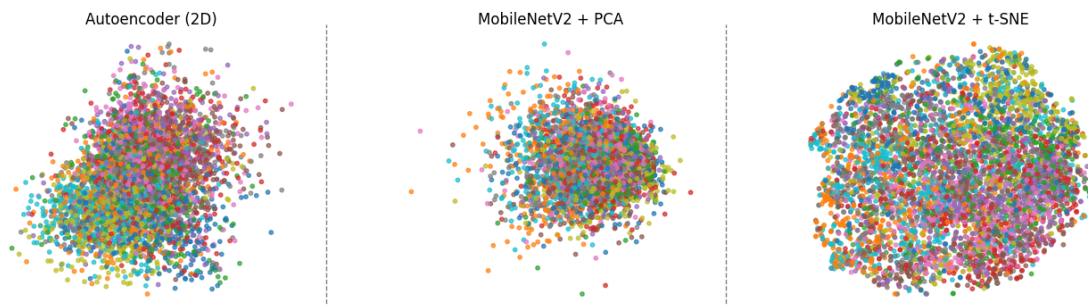


Figure 2: t-SNE, PCA comparison: MobileNetV2 (right) vs Autoencoder (left)

3. Denoising Autoencoder Training

We trained a denoising autoencoder using CIFAR-10 images corrupted with Gaussian noise. The model successfully reconstructed denoised outputs, preserving semantic content while removing noise.

Observation: The model achieved strong visual quality in reconstructions, even though pixel-wise accuracy was not evaluated numerically. The recovered images were noise-free, but a little blur issue was encountered.

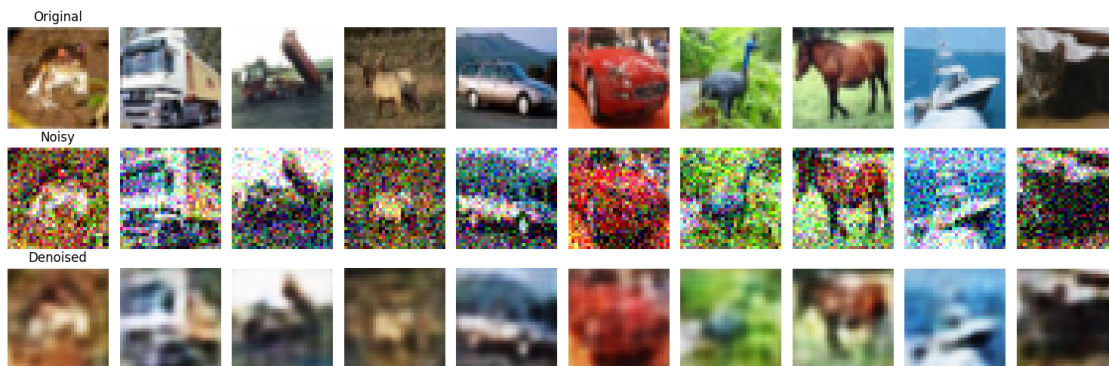


Figure 3: Original, Noisy, and Denoised CIFAR-10 images

4. CNN Classifier Without Augmentation

We trained a MobileNetV2-based classifier on 2000 CIFAR-10 images without any augmentation. After 20 epochs:

- Training accuracy reached **91.32%**
- Validation accuracy peaked around **73.75%**

Analysis: The model showed signs of overfitting as the gap between training and validation accuracy increased over epochs.

5. CNN Classifier With Data Augmentation

We evaluated two augmentation strategies using ImageDataGenerator:

- **Single Augmentation:** Applied only horizontal flipping.
- **Multiple Augmentation:** Applied flipping, rotation, shift, and zoom.

Results Overview

- **Single Augmentation:** Validation accuracy reached up to **74.25%**. Training accuracy peaked around **88.5%**, showing improved generalization compared to no augmentation.
- **Multiple Augmentation:** Validation accuracy reached up to **73.5%**, with training accuracy around **71%**. It provided stronger regularization, reducing overfitting further.

Conclusion

Both augmentation strategies improved generalization. However, single augmentation achieved slightly better validation accuracy, while multiple augmentations were more robust in reducing overfitting. **Visualization:**

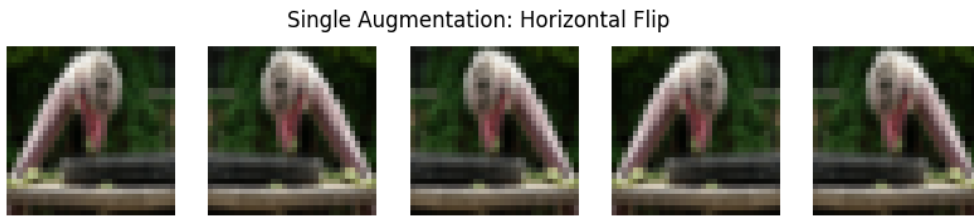


Figure 4: Single Augmentation (Horizontal Flip)

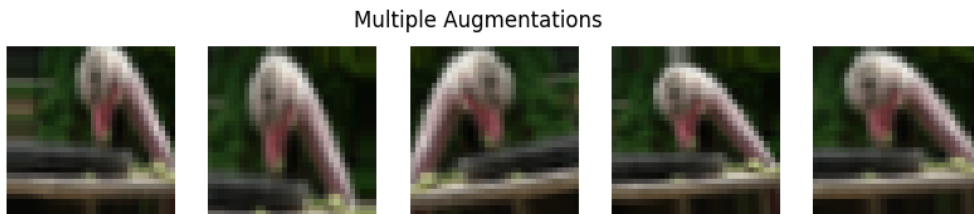


Figure 5: Multiple Augmentations (Flip + Rotate + Shift + Zoom)

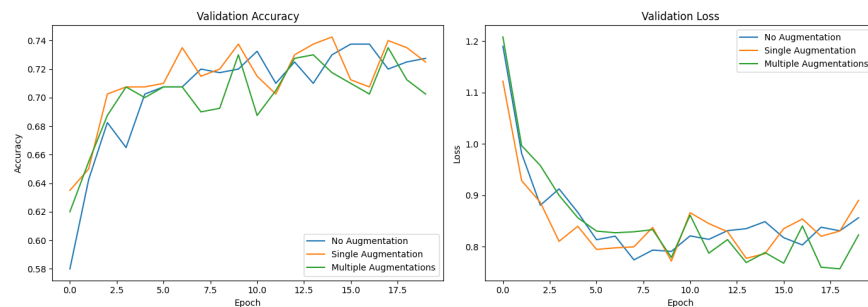


Figure 6: Accuracy trend

Code Link:

Here's the github link of the implemented code: <https://github.com/mdsajjadhossain25/CSE4261>